

Graph-Augmented Retrieval-Augmented Generation for Biomedical Literature Mining: A PubMed-Based Prototype for Breast Cancer Drug Delivery Evidence Retrieval

Prasanna Bala

OncoGraphRAG Project — Biomedical GraphRAG Literature Mining Prototype

Manuscript Year: 2026

Keywords

Biomedical NLP · GraphRAG · Retrieval-Augmented Generation · PubMed · Breast Cancer · Knowledge Graph · FAISS · Gemma · Drug Delivery · Nanoparticles · Oncology

Corresponding Author: Prasanna Bala. Project: OncoGraphRAG.

Received: 2026 | Revised: [date] | Accepted: [date]

Abstract

Background: Biomedical research literature is expanding at an unprecedented pace, with PubMed indexing over 36 million abstracts spanning molecular biology, oncology, pharmacology, drug delivery, and translational medicine. This proliferation makes it increasingly difficult for researchers to identify mechanistic relationships among diseases, therapies, drugs, biomarkers, genes, cell types, and drug-delivery platforms. Traditional retrieval systems return document lists but do not expose structured biomedical relationships or generate evidence-grounded, citation-traceable answers.

Objective: This study presents OncoGraphRAG, a lightweight Graph-Augmented Retrieval-Augmented Generation (GraphRAG) framework for PubMed-derived oncology literature mining, with breast cancer and nanoparticle-based drug delivery as a primary case study. The system is designed to be locally executable, modular, and extensible to larger biomedical corpora and higher-quality NLP components.

Methods: PubMed records were retrieved via the NCBI Entrez API and converted into 800-character overlapping text chunks. Dense embeddings were generated using BAAI/bge-base-en-v1.5 (768 dimensions) and indexed via FAISS IndexFlatIP. Biomedical entities were extracted using a dictionary-guided recognizer across eight semantic categories. A sentence-level graph construction module generated typed triples using ten relation rules. At query time, vector and graph retrieval were fused into an evidence context for a Gemma instruction-tuned language model, with a fallback module for insufficient model outputs.

Results: The prototype indexed 28,638 PubMed-derived chunks and constructed 7,695 sentence-level typed graph triples. End-to-end query execution successfully retrieved nanoparticle drug delivery evidence and generated a biologically plausible answer. Inline PMID citation enforcement and relation specificity require further improvement.

Conclusion: OncoGraphRAG demonstrates the feasibility of a locally executable biomedical literature mining system integrating dense retrieval, entity extraction, knowledge graph construction, and language model generation within a modular, reproducible pipeline. The prototype is extensible with transformer-based NER, UMLS normalization, Neo4j storage, and cross-encoder reranking.

1. Introduction

Biomedical literature has grown to a scale that challenges manual evidence synthesis. PubMed indexes over 36 million abstracts spanning molecular biology, oncology, pharmacology, and translational medicine [1]. Within breast cancer alone, literature covers multiple receptor subtypes, drug resistance mechanisms, nanoparticle delivery systems, and evolving targeted therapies. Conventional keyword search retrieves document lists but does not extract structured biomedical relationships or generate synthesized, citation-traceable answers aligned with specific research queries.

Retrieval-Augmented Generation (RAG) has emerged as a practical approach to grounding large language model (LLM) outputs in external evidence, substantially reducing hallucination relative to purely parametric generation [2]. In a standard RAG pipeline, a query is encoded into a dense vector representation, semantically similar passages are retrieved from an indexed corpus, and these passages are passed to an LLM for answer synthesis. However, vector-only RAG does not explicitly represent typed entity relationships. Two abstracts may share similar semantic distributions while describing contradictory drug mechanisms or different disease subtypes. Knowledge graphs complement dense retrieval by encoding typed relations such as DRUG—DELIVERED_VIA—THERAPY or GENE—INVOLVED_IN—MECHANISM, enabling discrimination between mechanistically distinct passages [3].

Breast cancer provides a well-motivated domain for GraphRAG research. The disease encompasses multiple molecular subtypes—hormone receptor-positive, HER2-amplified, and triple-negative—each with distinct therapeutic implications. Nanoparticle-based drug delivery has emerged as an active research area for improving therapeutic index, overcoming efflux-pump-mediated resistance, and enabling tumor microenvironment-targeted therapy [4, 5]. Literature also features well-characterized biomarkers (Oncotype DX, MammaPrint), targeted therapies (trastuzumab, pertuzumab), and resistance mechanisms (cancer stem cell plasticity, endocrine therapy resistance).

This paper presents OncoGraphRAG, a prototype biomedical GraphRAG system combining a FAISS dense vector index, dictionary-guided biomedical entity extractor, sentence-level knowledge graph, and Gemma instruction-tuned language model into a locally executable, end-to-end pipeline. Primary contributions include: (i) a complete reproducible prototype over 28,638 PubMed chunks and 7,695 typed graph triples; (ii) a sentence-level relation extraction module with ten typed biomedical relation categories; (iii) a hybrid evidence fusion architecture for LLM-based answer generation; and (iv) a structured evaluation plan with defined metrics and a 100-question benchmark.

2. Related Work

2.1 Retrieval-Augmented Generation in Biomedicine

The RAG paradigm, formalized by Lewis et al. [2], demonstrated that conditioning LLMs on dynamically retrieved passages substantially improves factual accuracy for knowledge-intensive tasks. Subsequent biomedical extensions—including BioRAG, PubMedQA-based systems [6], Almanac [7], and Med-PaLM 2 [8]—showed that domain-specific retrievers and LLMs outperform general models on clinical and molecular biology questions. However, most systems rely exclusively on dense vector retrieval and do not explicitly encode structured entity relationships between retrieved passages.

2.2 Biomedical Knowledge Graphs and GraphRAG

Curated biomedical knowledge graphs such as UMLS [9], PrimeKG [10], and DRKG [11] provide expert-validated entity-relation networks but require substantial annotation effort and cannot auto-update with new literature. Literature-driven graph construction systems such as SemRep [12] automate triple extraction from biomedical text at lower precision. The GraphRAG paradigm [3] combines vector retrieval with graph evidence; MedGraphRAG [13] demonstrated improvement over vector-only RAG on clinical benchmark datasets. The present work differs in constructing a domain-specific graph from PubMed sentence co-occurrence rather than curated resources, applied to a focused oncology use case.

2.3 Biomedical NER

Transformer-based NER models including BioBERT [14], PubMedBERT [15], and scispaCy [16] achieve strong performance on benchmarks such as NCBI Disease, BC5CDR, and JNLPBA. The present work employs a dictionary-guided NER module as a computationally efficient baseline, with transformer-based NER identified as the primary future improvement direction.

3. Materials and Methods

3.1 Data Source and Corpus Construction

PubMed records were retrieved via the NCBI Biopython Entrez API using an oncology-focused query covering cancer, carcinoma, tumor, and neoplasm terms, with downstream emphasis on breast cancer, nanoparticle drug delivery, therapy resistance, and molecular biomarkers. Each record preserved PMID, title, journal, year, DOI, PubMed URL, and abstract text. The corpus was stored in JSONL, JSON, and CSV formats to support independent inspection at each pipeline stage.

3.2 Chunking and Vector Indexing

Abstracts were segmented using a character-level sliding window with 800-character chunks and 100-character overlap, yielding 28,638 chunks each retaining source citation metadata. Character-level chunking was preferred over sentence-level segmentation to avoid fragmentation of long compound sentences common in biomedical abstracts. The 800-character window accommodates three to five biomedical sentences while remaining within the 512-token context limit of the embedding model. Dense embeddings (768-dimensional) were generated using BAAI/bge-base-en-v1.5 and L2-normalized for

cosine equivalence. Normalized embeddings were indexed with FAISS IndexFlatIP for exact nearest-neighbor retrieval.

Table 1. Corpus, chunking, and vector index configuration.

Parameter	Value
PubMed query scope	Cancer, oncology, breast cancer, nanoparticle drug delivery, therapy resistance
Chunk size / overlap	800 characters / 100 characters
Total chunks indexed	28,638
Embedding model	BAAI/bge-base-en-v1.5 (768-dimensional)
Normalization	L2 normalization — cosine similarity via inner product
Vector index	FAISS IndexFlatIP (exact brute-force nearest-neighbor)
Metadata fields	PMID, title, journal, year, DOI, PubMed URL, chunk ID
FAISS index file	bio_manual.faiss

BGE: BAAI General Embedding; FAISS: Facebook AI Similarity Search; IndexFlatIP: exact flat inner-product index.

3.3 Biomedical Entity Extraction

A dictionary-guided, rule-based NER module extracted entities across eight semantic categories. The entity lexicon covered full-form names and common abbreviations, with normalization mappings applied to standardize short forms (e.g., TNBC → triple-negative breast cancer; ADC → antibody-drug conjugate; EMT → epithelial-mesenchymal transition). Matching was performed after lowercasing and whitespace normalization using a longest-match priority strategy. Entities were stored with surface form, canonical form, type, chunk ID, PMID, and source sentence.

Table 2. Biomedical entity categories and representative examples.

Entity Type	Semantic Scope	Representative Entities
DISEASE	Cancer subtypes, disease states, pathological conditions	breast cancer; TNBC; metastatic breast cancer; HER2-positive; carcinoma in situ
THERAPY	Treatment modalities, delivery platforms, strategies	chemotherapy; immunotherapy; endocrine therapy; nanoparticles; liposomal encapsulation
DRUG	Specific pharmaceutical agents and compounds	doxorubicin; paclitaxel; trastuzumab; tamoxifen; pembrolizumab
GENE_PROTEIN	Genes, proteins, receptors, molecular targets	BRCA1; BRCA2; HER2/ERBB2; TP53; PD-L1; VEGF; Ki-67
BIOMARKER	Diagnostic and prognostic molecular indicators	recurrence score; Ki-67 index; HER2 status; hormone receptor positivity
ASSAY_TEST	Clinical, molecular, and genomic assays	Oncotype DX; MammaPrint; IHC; FISH; NGS
MECHANISM	Biological processes, pathways, resistance mechanisms	drug resistance; EMT; apoptosis; angiogenesis; metastasis; DNA repair
CELL_TYPE	Cellular populations and	breast cancer stem cells; tumor-infiltrating lymphocytes;

Entity Type	Semantic Scope	Representative Entities
	tumor microenvironment	macrophages; CD8+ T cells

TNBC: triple-negative breast cancer; IHC: immunohistochemistry; FISH: fluorescence in situ hybridization; NGS: next-generation sequencing.

3.4 Knowledge Graph Construction

Graph triples were constructed via sentence-level entity co-occurrence. Entities co-occurring within the same sentence were paired and passed through a priority-ordered relation inference module applying ten typed rules. Unmatched pairs received the ASSOCIATED_WITH fallback (LOW confidence). Each triple preserved source PMID, evidence sentence, confidence level, and full citation metadata. The graph triple schema included: src_text, src_canonical, src_type, relation, tgt_text, tgt_canonical, tgt_type, confidence, pmid, title, journal, year, pubmed_url, chunk_id, and evidence sentence.

Table 3. Biomedical relation type catalogue.

Relation	Source → Target	Conf.	Example Triple
TREATS	THERAPY/DRUG → DISEASE	H	paclitaxel —TREATS→ triple-negative breast cancer
TARGETS	THERAPY/DRUG → CELL_TYPE/GENE	H	trastuzumab —TARGETS→ HER2/ERBB2
DRIVES	CELL_TYPE → MECHANISM	H	cancer stem cells —DRIVES→ drug resistance
CAUSES_RESISTANCE	MECHANISM → THERAPY/DRUG	M	EMT —CAUSES_RESISTANCE→ endocrine therapy
DELIVERED_VIA	DRUG → THERAPY	H	doxorubicin —DELIVERED_VIA→ nanoparticles
INDICATES	BIOMARKER → DISEASE	H	HER2 status —INDICATES→ HER2+ breast cancer
MEASURES	ASSAY/TEST → BIOMARKER/GENE	H	Oncotype DX —MEASURES→ recurrence score
EXPRESSED_BY	GENE/PROTEIN → CELL_TYPE	M	PD-L1 —EXPRESSED_BY→ tumor-infiltrating lymphocytes
INVOLVED_IN	GENE/PROTEIN → MECHANISM	H	BRCA1 —INVOLVED_IN→ DNA repair
INFLUENCES	MECHANISM → DISEASE	M	angiogenesis —INFLUENCES→ metastatic breast cancer
ASSOCIATED_WITH	Any → Any	L	nanoparticles —ASSOCIATED_WITH→ drug delivery

H = HIGH, M = MEDIUM, L = LOW confidence. ASSOCIATED_WITH is applied when no specific rule matches.

3.5 GraphRAG Query Pipeline

At query time, the pipeline: (1) encodes the query using BGE; (2) retrieves top-k FAISS vector chunks by inner product score; (3) extracts biomedical query terms for graph matching; (4) scores graph triples by query-term overlap and provenance overlap with retrieved chunks; (5) fuses vector chunks and high-

confidence graph triples into a structured evidence prompt; (6) generates an answer via Gemma-3-1B-it conditioned on the fused evidence. A fallback module returns a structured evidence-based answer when model output is insufficient—empty, truncated, or off-topic.

4. System Architecture

Offline Indexing Pipeline	Online Query Pipeline
PubMed Entrez Query ↓ 01_download_pubmed.py JSONL Corpus ↓ 02_chunk_articles.py 28,638 Chunks + Metadata ↓ 03_build_vector_index.py BGE Embeddings → FAISS Index ↓ 05_extract_entities.py Biomedical Entities ↓ 06_build_graph_triples.py 7,695 Graph Triples (CSV/JSONL)	User Question ↓ BGE Query Encoder ↳ FAISS Top-K Retrieval PubMed Evidence Chunks ↳ Graph Triple Scoring Typed Relation Evidence ↓ Evidence Fusion Module Structured Evidence Prompt ↓ Gemma-3-1B-it Citation-Aware Answer + JSON

5. Implementation

The system was implemented as a modular Python pipeline organized into eight sequentially numbered scripts. Each script reads from explicit input files and writes to explicit output files, enabling independent inspection, debugging, and partial re-execution of any stage without rerunning the full pipeline.

Table 4. Implementation modules and hardware configuration.

Component	Detail
01_download_pubmed.py	PubMed corpus acquisition via Biopython Entrez API; stores JSONL, JSON, CSV
02_chunk_articles.py	800-character overlapping chunking with citation metadata preservation
03_build_vector_index.py	BGE embedding generation and FAISS IndexFlatIP construction
04_test_vector_search.py	Semantic retrieval validation and metadata alignment verification
05_extract_entities.py	Dictionary-guided biomedical NER with abbreviation normalization
06_build_graph_triples.py	Sentence-level entity pairing, relation inference, provenance storage
07_graphrag_query.py	Hybrid vector + graph retrieval and evidence fusion
generate_with_gemma.py	Gemma-3-1B-it answer generation and fallback module; JSON/text output
GPU / OS	NVIDIA RTX 3050 4GB VRAM · Windows 11 · PyCharm · Python 3.10 venv
Core libraries	FAISS 1.7 · Hugging Face Transformers 4.x · SentenceTransformers · PyTorch 2.x

GPU: graphics processing unit; VRAM: video random access memory.

6. Results

6.1 System Output Summary

Table 5. Prototype quantitative output summary.

Metric	Value	Status
PubMed chunks indexed	28,638	Completed
Embedding dimension	768 (BGE-base)	Completed
Graph triples generated	7,695	Completed
Entity categories	8	Completed
Relation types	10	Completed
End-to-end query execution	Passed (test query)	Completed
Fallback answer generation	Implemented	Completed
Inline PMID citation	Partial — enforcement needed	Needs improvement
Relation specificity	ASSOCIATED_WITH fallback in subset	Needs improvement

6.2 Example Query Execution

Test query: "How are nanoparticles used in breast cancer drug delivery?" The system completed all pipeline stages — query embedding, FAISS top-k retrieval, graph triple scoring, evidence fusion, and Gemma generation. The top retrieved PMID was 41501437, covering precision nanotechnology and drug-resistant breast cancer. Graph evidence included DELIVERED_VIA and TARGETS triples not ranked in the top vector results by cosine similarity, demonstrating graph complementarity to dense retrieval.

Table 6. Example query execution summary.

Field	Output
Query	How are nanoparticles used in breast cancer drug delivery?
Top PMID retrieved	41501437
Top topic	Precision nanotechnology and drug-resistant breast cancer
Graph relations surfaced	DELIVERED_VIA, TARGETS, ASSOCIATED_WITH
Generation model	google/gemma-3-1b-it
Citation behavior	PMIDs in context; per-sentence attribution requires enforcement

PMID: PubMed Identifier.

6.3 Generated Answer

The Gemma 1B model generated the following answer during preliminary testing:

Generated Answer (Gemma-3-1B-it, verbatim)

Nanoparticles offer a promising avenue for enhancing breast cancer drug delivery due to their ability to overcome challenges associated with conventional therapies. Specifically, nanoparticles can target cancer stem cells (CSCs), circumventing efflux pumps and modulating the immune response, ultimately improving treatment efficacy. Studies demonstrate their effectiveness in delivering drugs directly to tumors, inhibiting metastasis, and synergistically combining with immunotherapy and chemotherapy. Furthermore, research indicates that nanoparticles can be designed to release therapeutic agents in a controlled manner, influencing the tumor microenvironment and potentially enhancing drug delivery. While challenges remain regarding stability and immunogenicity, ongoing research continues to refine nanoparticle formulations for improved clinical translation.

The response covers key mechanisms including efflux pump circumvention, tumor microenvironment modulation, controlled drug release, and combination therapy—biologically plausible given the retrieved PubMed evidence. However, per-sentence PMID attribution is absent, attributed to the instruction-following limits of the 1B-parameter model.

6.4 Evidence Grounding Assessment

Table 7. Preliminary evidence grounding assessment.

Criterion	Observation	Assessment
PubMed-based retrieval	All chunks derived from PubMed corpus	Pass
Vector search execution	FAISS retrieval completed successfully	Pass
Graph search execution	Triple scoring and retrieval completed	Pass
LLM generation	Gemma produced coherent biomedical answer	Pass
Inline PMID citation	PMIDs in context but not embedded per-claim	Needs improvement
Relation specificity	ASSOCIATED_WITH fallback used for subset of triples	Needs improvement
Clinical safety boundary	No clinical dosing or treatment instructions generated	Pass
Reproducibility	All artifacts saved in JSONL, CSV, FAISS formats	Pass
Evidence traceability	Each chunk and triple linked to source PMID	Pass

7. Discussion

7.1 Evidence Traceability and Transparency

A core design principle of OncoGraphRAG is evidence traceability. Every retrieved chunk links to a PubMed identifier, enabling the end-user to verify any retrieved evidence claim. Each graph triple additionally preserves the verbatim evidence sentence and PMID, supporting independent assessment of relation quality. This transparency distinguishes the system from closed-box LLM generation and aligns with requirements for evidence-grounded biomedical AI systems.

7.2 Hybrid Retrieval Complementarity

The combination of dense vector retrieval and structured graph retrieval addresses complementary failure modes. Dense retrieval identifies topically relevant passages but may conflate mechanistically distinct topics. Graph retrieval explicitly encodes typed relations, surfacing DELIVERED_VIA and TARGETS triples that were not top-ranked by cosine similarity alone in the nanoparticle test query. This complementarity motivates the GraphRAG architecture in biomedical settings where mechanistic specificity matters.

7.3 Language Model Limitations

The most significant qualitative limitation is Gemma 1B's inconsistent per-claim citation formatting. This reflects the instruction-following limits of a 1 billion parameter model—sufficient for coherent text generation but insufficient for reliable structured citation output without explicit fine-tuning. Two improvement paths are identified: (1) prompt engineering with explicit per-claim citation templates, and (2) evaluation of larger models including Gemma 4B, MedGemma, and BioMistral with domain-specific instruction tuning.

7.4 Graph Coverage and Scalability

The 7,695 triples generated from 28,638 chunks yield approximately 0.27 triples per chunk, reflecting the selectivity of sentence-level co-occurrence extraction. The high ASSOCIATED_WITH fallback rate reduces discrimination value and will be addressed by a supervised biomedical relation classifier trained on ChemProt and BioRED. For corpus scaling beyond 100,000 chunks, the current flat FAISS index will require transition to approximate nearest-neighbor indexes (IndexIVFFlat or HNSW) for sub-linear query latency.

8. Limitations

Table 8. Current system limitations and planned improvements.

Limitation	Impact	Planned Improvement
Abstract-only corpus	Misses detailed mechanisms in full-text articles	Integrate PMC open-access full-text
Dictionary-guided NER	Misses novel and low-frequency biomedical entities	Add scispaCy, BioBERT, or PubMedBERT NER
Rule-based relation extraction	High ASSOCIATED_WITH fallback rate; low discrimination	Supervised classifier on ChemProt / BioRED
Gemma 1B model	Inconsistent per-claim PMID citation formatting	Evaluate Gemma 4B, MedGemma, BioMistral 7B
No formal benchmark	Cannot quantify retrieval or generation quality	100-question annotated evaluation set
No expert review	Cannot claim research-grade answer quality	Biomedical expert annotation of outputs
Flat FAISS index	Linear query latency — impractical beyond 100k chunks	Transition to IndexIVFFlat or HNSW

Limitation	Impact	Planned Improvement
No UMLS normalization	Synonym variants produce duplicate graph nodes	Integrate QuickUMLS for CUI-level deduplication
No baseline comparison	Cannot quantify GraphRAG gain over simpler systems	Compare against keyword, vector-only, graph-only RAG

NER: named entity recognition; UMLS: Unified Medical Language System; CUI: concept unique identifier.

9. Evaluation Plan

A formal evaluation will use a manually curated benchmark of 100 biomedical literature questions across seven question categories representative of the breast cancer oncology domain. Table 9 details the benchmark and Table 10 defines evaluation metrics. Table 11 outlines the planned baseline comparison to quantify GraphRAG improvement over simpler retrieval approaches.

Table 9. Proposed 100-question evaluation benchmark.

Category	Example Question	N
Nanoparticle drug delivery	How are nanoparticles used in breast cancer drug delivery?	20
Therapy resistance	What mechanisms drive endocrine therapy resistance in breast cancer?	20
Cancer stem cells	What therapies selectively target breast cancer stem cells?	15
Prognostic biomarkers	Which biomarkers predict breast cancer recurrence?	15
Diagnostic assays	What is the clinical role of Oncotype DX in early-stage breast cancer?	10
Molecular mechanisms	Which genes are associated with breast cancer metastasis?	10
Combination therapy	How are immunotherapy and chemotherapy combined in TNBC?	10
Total	—	100

TNBC: triple-negative breast cancer.

Table 10. Proposed evaluation metrics.

Metric	Definition
Recall@5 / Recall@10	At least one relevant PubMed chunk in top 5 / top 10 FAISS results
Citation accuracy	Fraction of cited PMIDs in the generated answer that verifiably support the claim
Answer faithfulness	Fraction of generated claims grounded in retrieved evidence
Graph relation relevance	Fraction of retrieved triples topically related to the query
Hallucination rate	Fraction of generated claims with no support in retrieved evidence
Expert usefulness score	Biomedical expert rating 1 (not useful) to 5 (clinically useful)

Recall@K: fraction of queries for which at least one relevant chunk appears in the top-K results.

Table 11. Planned baseline and ablation comparison systems.

System	Description
Keyword PubMed search	Standard PubMed keyword query — no semantic retrieval or graph augmentation
Vector-only RAG	FAISS semantic retrieval without graph triples; Gemma from vector context only
Graph-only retrieval	Graph triple retrieval without vector chunks; Gemma from graph context only
GraphRAG (proposed)	Combined FAISS vector + graph triple retrieval with evidence fusion
GraphRAG + BM25 hybrid	GraphRAG with additional BM25 sparse retrieval via reciprocal rank fusion
GraphRAG + reranker	GraphRAG with cross-encoder reranking of fused candidate evidence
GraphRAG + transformer NER	GraphRAG with scispaCy or BiomedBERT replacing dictionary NER

10. Future Work

The OncoGraphRAG prototype establishes a reproducible baseline from which targeted component improvements can be systematically evaluated. The modular pipeline design facilitates independent replacement of any stage without disrupting the rest of the pipeline, enabling rigorous ablation studies across NER, relation extraction, retrieval, and generation components.

Future development targets five primary directions. First, transformer-based NER—integrating scispaCy, BioBERT, or PubMedBERT with UMLS QuickUMLS normalization for CUI-level entity deduplication. Second, supervised relation extraction—replacing rule-based inference with a classifier trained on ChemProt and BioRED to reduce ASSOCIATED_WITH fallback rate. Third, hybrid retrieval—combining FAISS dense retrieval with BM25 sparse search using reciprocal rank fusion and cross-encoder reranking for improved precision. Fourth, Neo4j graph storage—migrating from flat JSONL/CSV triple stores to a Neo4j graph database enabling multi-hop Cypher traversal and LangChain integration via Neo4jGraph and Neo4jVector. Fifth, model comparison—evaluating Gemma 4B, MedGemma, BioMistral 7B, Qwen2.5-7B, and Llama-3-8B-Instruct on answer faithfulness, citation accuracy, hallucination rate, and expert usefulness score.

11. Conclusion

This paper presents OncoGraphRAG, a locally executable, modular Graph-Augmented Retrieval-Augmented Generation framework for PubMed-derived oncology literature mining. The completed prototype indexes 28,638 biomedical text chunks, constructs a knowledge graph of 7,695 sentence-level typed triples across ten relation categories, and integrates FAISS dense retrieval, graph triple retrieval, evidence fusion, and Gemma-3-1B instruction-tuned generation into a reproducible end-to-end pipeline.

Preliminary evaluation on a nanoparticle breast cancer drug delivery query demonstrates successful vector retrieval, relation-specific graph evidence surfacing, and coherent biomedical answer generation with full evidence traceability to source PMIDs. Primary limitations—rule-based NER, high fallback relation rate, and Gemma 1B citation inconsistency—are addressed by a defined roadmap encompassing transformer-based NER, supervised relation extraction, hybrid retrieval, Neo4j graph storage, and larger biomedical language models.

As biomedical literature continues to expand at scale, systems capable of structured evidence retrieval, relation extraction, and citation-traceable answer generation become increasingly important for research synthesis and clinical knowledge translation. OncoGraphRAG provides a reproducible prototype foundation for this class of systems in the oncology domain.

Ethics Statement

This system is intended solely for biomedical literature mining and research assistance. It does not interact with patient data, clinical records, or personal health information. All text was derived from publicly available PubMed abstracts. The system is not intended for clinical diagnosis, treatment selection, or patient-specific medical decision support. All outputs should be reviewed by qualified biomedical domain experts before research or clinical application.

Data and Code Availability

The OncoGraphRAG pipeline and all intermediate artifacts (FAISS index, chunk metadata, entity extraction outputs, graph triples, query results) are available upon request to support replication. PubMed abstracts were accessed via the NCBI Entrez API under NCBI data usage policies. Code will be made available in a public repository upon manuscript acceptance.

Appendix: Pipeline Output Artifacts

Table A1. Complete list of OncoGraphRAG pipeline output artifacts.

File	Stage	Format	Description
cancer_pubmed_10k.jsonl	Corpus	JSONL	Raw PubMed corpus with abstract text and citation metadata
bio_chunks.jsonl	Chunking	JSONL	Overlapping 800-character chunks with chunk IDs
bio_chunk_metadata.json	Chunking	JSON	Metadata mapping for all 28,638 chunks

File	Stage	Format	Description
bio_manual.faiss	Indexing	Binary	FAISS IndexFlatIP over 768-dim BGE embeddings
bio_entities.jsonl	NER	JSONL	Per-chunk extracted biomedical entities with type and canonical form
bio_entity_summary.csv	NER	CSV	Entity frequency aggregated by semantic type
bio_graph_triples.csv	Graph	CSV	All 7,695 typed graph triples in tabular format
bio_graph_triples.jsonl	Graph	JSONL	All 7,695 triples in streaming JSON Lines format
bio_graph_summary.csv	Graph	CSV	Relation frequency and entity co-occurrence statistics
graph_construction_stats.json	Graph	JSON	Graph diagnostic statistics
graphrag_result_*.json	Query	JSON	Per-query results including vector and graph evidence
gemma_answer_*.json	Generation	JSON	Final generated answer with evidence and metadata
gemma_answer_*.txt	Generation	Text	Plain text final answer for human review

Author Contributions

Prasanna Bala: Conceptualization, system architecture design, full pipeline implementation (corpus acquisition, chunking, embedding, entity extraction, graph construction, query pipeline, answer generation), analysis of results, and manuscript preparation.

Acknowledgements

The author acknowledges the following open-source tools and resources used in this prototype: the National Center for Biotechnology Information (NCBI) PubMed and Entrez API for literature access; Facebook AI Research (FAIR) for the FAISS vector search library; BAAI for the BGE-base-en-v1.5 embedding model; Google for the Gemma-3-1B-it instruction-tuned language model; and Hugging Face for the Transformers and SentenceTransformers libraries. PubMed literature was accessed in accordance with NCBI data usage policies.

References

- [1] National Library of Medicine. PubMed Overview. NCBI. 2026. <https://pubmed.ncbi.nlm.nih.gov/about/>
- [2] Lewis P, et al. Retrieval-augmented generation for knowledge-intensive NLP tasks. NeurIPS. 2020;33:9459–74.

- [3] Edge D, et al. From local to global: a graph RAG approach to query-focused summarization. arXiv:2404.16130. 2024.
- [4] Blanco E, Shen H, Ferrari M. Principles of nanoparticle design for overcoming biological barriers to drug delivery. Nat Biotechnol. 2015;33:941–51.
- [5] Meng H, et al. Engineered design of mesoporous silica nanoparticles to deliver doxorubicin and P-gp siRNA to overcome drug resistance. ACS Nano. 2013;7:994–1005.
- [6] Jin Q, et al. PubMedQA: A dataset for biomedical research question answering. EMNLP. 2019.
- [7] Zakka C, et al. Almanac — Retrieval-augmented language models for clinical medicine. NEJM AI. 2024;1(2).
- [8] Singhal T, et al. Large language models encode clinical knowledge. Nature. 2023;620:172–80.
- [9] Bodenreider O. The Unified Medical Language System (UMLS). Nucleic Acids Res. 2004;32:D267–70.
- [10] Chandak P, Huang K, Zitnik M. Building a knowledge graph to enable precision medicine. Sci Data. 2023;10:67.
- [11] Ioannidis VN, et al. DRKG — Drug Repurposing Knowledge Graph for COVID-19. GitHub. 2020.
- [12] Rindflesch TC, Fiszman M. The interaction of domain knowledge and linguistic structure in NLP. J Biomed Inform. 2003;36:462–77.
- [13] Wu Y, et al. MedGraphRAG: A knowledge graph-enhanced RAG framework for medical QA. arXiv:2408.04187. 2024.
- [14] Lee J, et al. BioBERT: a pre-trained biomedical language representation model. Bioinformatics. 2020;36:1234–40.
- [15] Gu Y, et al. Domain-specific language model pretraining for biomedical NLP. ACM TCBB. 2022;3:1–23.
- [16] Neumann M, et al. ScispaCy: fast and robust models for biomedical NLP. BioNLP. 2019.